INORGANIC CUME
APRIL 16, 2004

There are a number of cyclic structures among the various combinations of the main group elements that have fascinated chemists for many decades.

A. Indicate the structure of the ring compound generated by the following reactions:

a. $nPCl_5 + nNH_4Cl \rightarrow$

b. $NaBH_4 + NH_4Cl \rightarrow$

c. $NaBH_4 + H_3PO_4 \rightarrow$

d. $NaH_2PO_4 + H_3PO_4 \rightarrow$

e. S_4N_4 + pass hot vapor over silver wool $\rightarrow$

f. Sulfur in standard state.

B. The Lewis Dot structures for many of these generated much interest in the bonding. Give the Lewis Dot structures for each and comment about potential delocalization.

C. At least one of these might be considered "aromatic". Indicate which one and why.

D. At least one of these has some significant commercial value. Indicate which one and why?

E. The product in "b" spontaneously inflames on contact with air. Why? Hint: Consider the products.

F. The product of "e" is extremely shock sensitive. Why? Hint: Consider the products